

CHARACTERIZATION OF A 500J DENSE PLASMA FOCUS FOR PRODUCING SOFT X-RAYS *

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Abstract

The scaling of x-ray output with (stored energy)² and ~10ns radiation pulse width of the dense plasma focus make it an interesting source of soft x-ray radiation for lithography, biological imaging, nano imaging or soft x-ray diagnostic calibration. AASC was funded by DTRA to explore soft x-ray diagnostic calibration*. In this follow-on study, we explore an alternative electrode configuration with Ar. The soft x-radiation (>1 keV) yield, radiation pulse width and debris are characterized for our 500J dense plasma focus over 100's of pulses fired at 0.2 Hz. The radiation yield is compared with (current)⁴ scaling. A soft x-ray spectrometer is installed to examine the soft x-ray spectrum.

I. INTRODUCTION

The dense plasma focus (DPF) is a pulsed radiation source relying on pinch physics to generate an intense radiation burst [1-4]. In this paper we examine a 500J DPF operating in Argon gas with a peak current of 140kA. The DPF was developed and diagnosed under a DTRA Ph I SBIR (HDTRA1-06-P-0097). The DPF was modified to calibrate an x-ray crystal spectrometer (XRCS). The XRCS needed to be calibrated for Aluminum lines from a high current z-pinch.. The Argon K-shell lines in second order lie close to the position of Al K-shell lines in first order, making Ar an excellent source for Aluminum. This paper outlines the measurement of: the Argon yield; the spatial dimensions of the pinch; the Argon x-ray spectrum.

II. EXPERIMENTAL CONFIGURATION

The DPF is driven by ten 600nF GA capacitors and ten P-E spark gap switches configured as ten parallel, pulsed power modules. The inductance of the circuit when shorted just outside the vacuum is 17nH. The capacitor bank stores 500J in this operating mode. The electrodes are SS304 and alumina silicate is used as the insulator for the preliminary characterization. Alumina (99.6%) was later used to reduce debris and improve insulator lifetime.

Alumina has proven to be superior to many other insulators for producing radiation. The anode has a diameter of 20mm for these experiments with eight 3.2mm cathode rods located on a 20mm radius circle (See Figure 1).

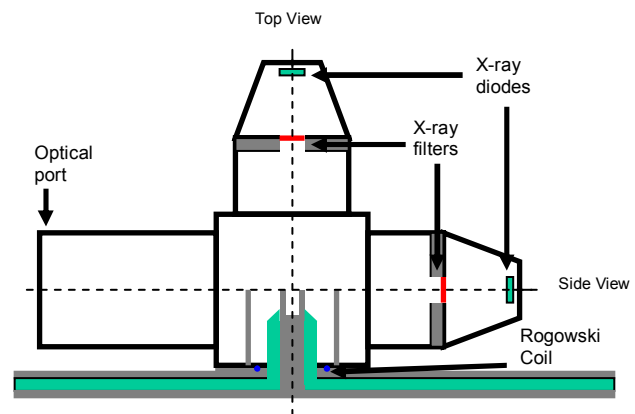


Figure 1: DPF configuration for these experiments.

The DPF can operate at up to 1Hz with the existing power supplies and data acquisition system (DAQ). The DAQ is a Tektronix TVS645A (5Gs/s) with a GPIB interface controlled by Labview. Data are typically digitized at 1Gs/s.

The DAQ digitized 4 channels during a normal run: the voltage at the load measured by a NorthStar 60kV, 100MHz H.V. probe; di/dt measured by a 3 turn Rogowski coil installed in the cathode plate; voltage near the capacitors measured by a Tektronix P6015A H.V. probe; soft x-radiation measured by a 1cm² Quantrad Si PIN diode. The diode was filtered with 12.7μm Kapton (1000Å Al), 25.4μm Aluminum or 12.7μm Titanium. Titanium filters were used for the most reliable K-shell yield estimates. The Kapton filters were used to look at the total spectrum. Aluminum filters were used to calculate the ratio of the exposed pinch length to the total pinch length. In some experiments, the Tektronix H.V. probe was replaced with a second 1cm² Quantrad Si PIN diode. In addition to the electrical signals, a 6.0Mpixel Kodak DCS760 camera was used with a 500μm aperture to record visible images of the pinch. The images were

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used to locate the pinch for alignment with the spectrometer.

The XRCS was side mounted on the DPF to minimize debris buildup on the 12.7 μ m Kapton (1000 $\text{\AA}$ Al) filter during a run. The spectrometer was fitted with a TIAP crystal. Initial calculations based upon I^4 scaling showed that >10,000 pulses would be required for regular film to achieve an N.D.=0.1 so the spectrometer was refitted with digital imaging plates (see Figure 2). A Fuji BAS-5000 scanner was used to read the plates. Only 1000 pulses would be required to achieve a measurable spectrum at 10mJ/cm per pulse predicted by I^4 scaling. The DPF is capable of operating for 4000 pulses per day so a single spectrum could be recorded per day when one considers pump out time.

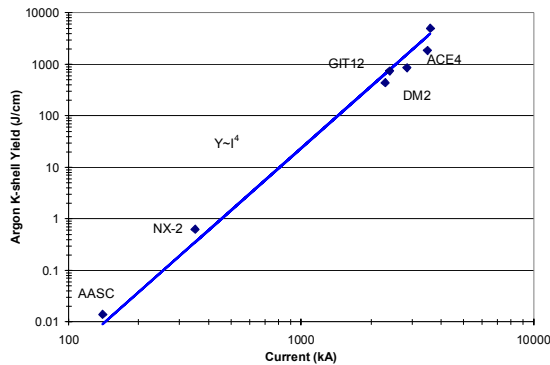


Figure 2: Radiation yield per unit pinch length vs. the drive current.

III. X-RAY YIELD, PULSE WIDTH AND REPRODUCIBILITY

X-rays are produced in a DPF when a magneto-sonic plasma driven by magnetic pressure implodes on the axis of the anode. The anode is frequently hollow to allow the pinch to form without producing substantial debris from the anode. The DPF has three steps leading to the implosion: breakdown/initiation; axial run-down; radial implosion. Figure 3 labels the breakdown on the voltage trace and the pinch on the current trace.

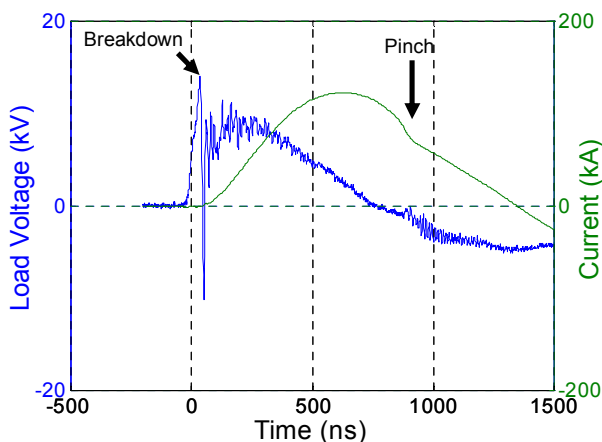


Figure 3: Voltage and current trace for typical DPF pinch.

The axial run down begins as the current rises and continues until the peak of the current. Two Quantrad Si PIN diodes were used to examine the radiation yield from the pinch. One diode was positioned directly above the pinch. The second diode viewed the pinch from the inlet to the XRCS. Both diodes were biased to -250V. Figure 4 shows the results of the integrated diode signals from 1000 pulses.

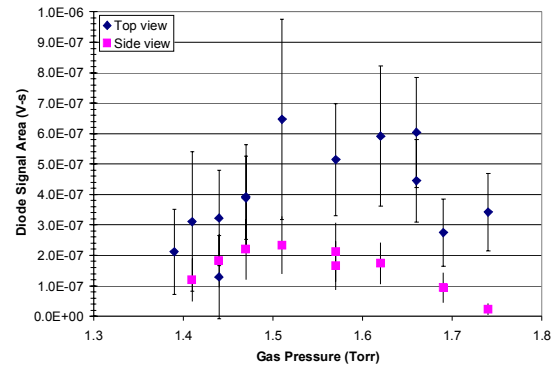


Figure 4: Radiation yield per pulse from the top view and side view

The optimum pressure in Argon occurs at 1.51 Torr from both the top view and side view. Both measurements have a standard deviation of ~50% for 1000 pulse experiments. Such a variation is known to occur for DPF systems that do not flow the working gas to maintain purity. Since we are interested in an integral measurement, we can tolerate the large variation. The ratio of top view to side view diode signals at the optimum pressure is 2.53 ± 0.65 . This suggests that a majority of the useful radiation normal to the pinch axis is deep within and absorbed by the anode.

The diode signal is converted to radiation dose using a 12.7 μ m Ti filter with data taken at the optimum pressure. The bulk of the line radiation is assumed to be concentrated at the He- α line. The K-shell yield is 12mJ when measured from the top view. Lee states that the pinch length is $\sim 0.7 \times$ the anode diameter, so we know the radiation per unit length is 8.6 ± 4.1 mJ/cm using an alumina silicate insulator. When an alumina insulator was installed, 10 ± 4.7 mJ/cm was achieved. The increase in x-ray yield is attributed to the alumina (99.6%) producing less debris during the breakdown phase.

Although the radiation pulse width is not directly relevant to the time integrated measurement on the XRCS, the duration of the pinch may be relevant to other applications. A typical Argon diode trace is shown in Figure 5. The Full Width at Half Maximum (FWHM) is 18ns.

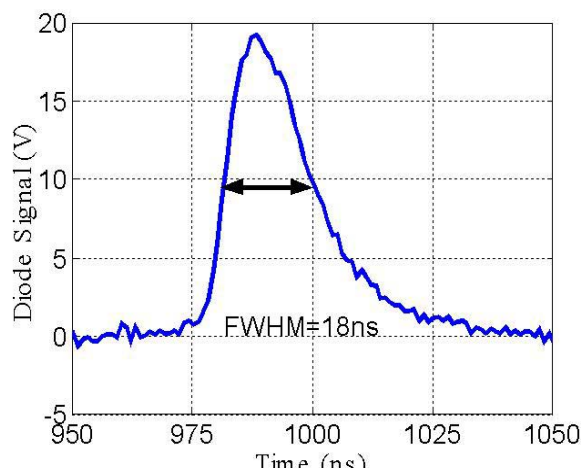


Figure 5: Argon X-radiation pulse from 500J DPF with an alumina silicate insulator

The choice of insulator did not seem to change the FWHM of the radiation pulse, but more detailed studies are required to better understand the difference in the two insulators.

The final study on the x-radiation production was the reproducibility of the source. As shown in Figure 6, the radiation yield tends to cluster but large radiation yield pulses persist throughout the experiment.

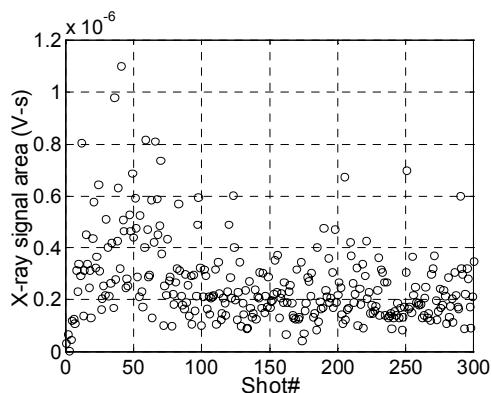


Figure 6: Typical 300 pulse experiment for the DPF in Argon

A longer experiment of 2000 pulses was also completed. The variation in radiation yield did not improve with the number of pulses. Prior work has shown that flowing gas and cooling the anode can reduce the standard deviation in the radiation yield from ~50% to 25%.

IV. VISIBLE PINCH IMAGES

Before installing the XRCS onto the DPF, it was important to ensure the XRCS could be view the pinch without any obstructions. The Kodak digital camera was configured to view the pinch from the side view with a 500 μ m aperture. The camera was configured for 1:1

magnification. A visible image of an Argon pinch is shown in Figure 7.

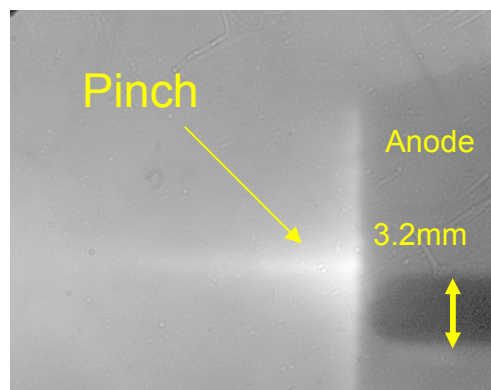


Figure 7: Visible image of an Argon pinch from the side view

The dark region 3.2 mm in height is the shadow of one of the cathode rods. The dimensions of the pinch were measured using this cathode rod as a reference. The FWHM of the pinch was an average over the visible pinch length in Figure 7. The FWHM of the pinch is 2.0mm from Figure 7. The analysis of other images showed that the pinch diameter only varied by 0.2mm.

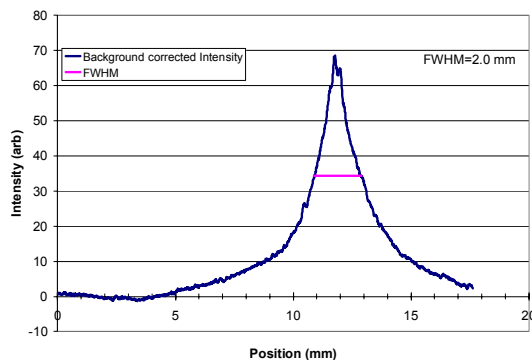


Figure 8: Vertical line out of Figure 7 over the pinch region.

A horizontal line out was used to determine the pinch length. The anode served as one end of the measurement. The other end of the measurement was taken to be 10% of the peak intensity. Figure 10 shows a pinch with a 5.2mm length. Using the x-ray yield ratio from top view and side view, this would mean the full pinch length is 13.2 ± 3.4 mm which is roughly consistent with Lee's findings that the pinch length is $0.7 \times$ the pinch diameter or 14mm. (Simply doubling the 5.2 mm length measured would be fully consistent with Lee's rule-of-thumb.)

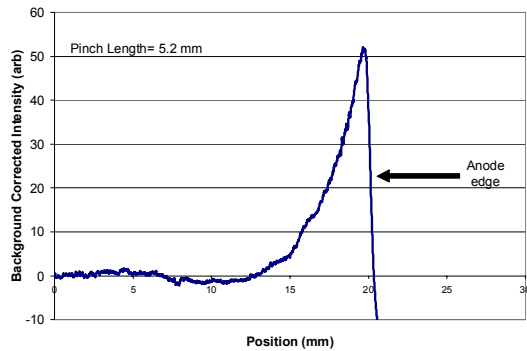


Figure 10: Horizontal line out of the Figure 7 along the pinch axis

V. X-RAY CRYSTAL SPECTROMETER

The XRCS was mounted to the side view diode port on the DPF. A Quanrad diode with a $12.7\mu\text{m}$ Ti filter viewed the pinch from above. An alumina insulator was installed. The cathode rods were shortened slightly to ensure a clear line of sight between the pinch and the crystal. A $12.7\mu\text{m}$ Kapton ($1000\text{\AA}$ Al) filter covered the 2.5 cm diameter aperture. The TIAP crystal was 50cm from the source. The long distance is to ensure the filters would survive the long runs. If the filter was damaged, debris could reach the crystal impairing its performance. A gate valve was installed between the crystal and the pinch to ensure the DIP was not exposed during warm up experiments. Every time the electrodes were exposed to air a 300 shot conditioning run was required to produce x-rays. The second order Argon spectrum for a 500J DPF is shown in Figure 11.

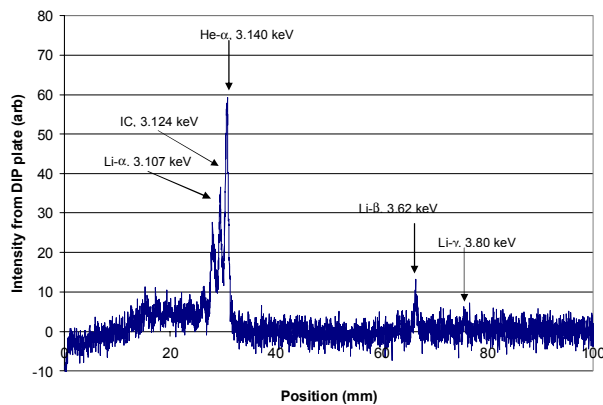


Figure 11: Line identification from second order Argon spectrum from 500J DPF.

As expected the Ar He- α line is the most intense line in the spectrum. The Ar Lyman- α line is absent which suggests the pinch temperature is less than 800eV. The spectrum required only 800 shots. Further analysis of the spectrum and other experiments are on-going.

VI. SUMMARY

A 500J dense plasma focus (DPF) was used to calibrate an x-ray crystal spectrograph (XRCS) using Argon K-shell radiation in second order. The XRCS was designed to measure K-shell radiation of Aluminum in first order so second order Argon is the best operating gas for the DPF. Before installing the spectrometer, the radiation yield per pulse was measured to be $10\pm 4.7\text{mJ/cm}$ of K-shell Argon assuming that the He- α line is the dominant radiator. Later measurements with XRCS confirmed our assumption. The x-ray pulse width (18ns) was consistent with larger pinch machines such as Saturn for K-shell Argon radiation. The reproducibility of the source was determined to be reasonable for a DPF but flowing gas and anode cooling are required to improve the radiation yield standard deviation from $\sim 50\%$ to 25%.

The pinch diameter and pinch length were measured using visible images from a digital camera. The pinch diameter was 2.0 mm (FWHM) and the exposed pinch length is 5.2mm.

With the XRCS installed, a clear spectrum with Argon He- α and Li features were clearly visible. The Lyman- α was missing from the spectrum which indicates the pinch temperature is less than 800eV.

VII. REFERENCES

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